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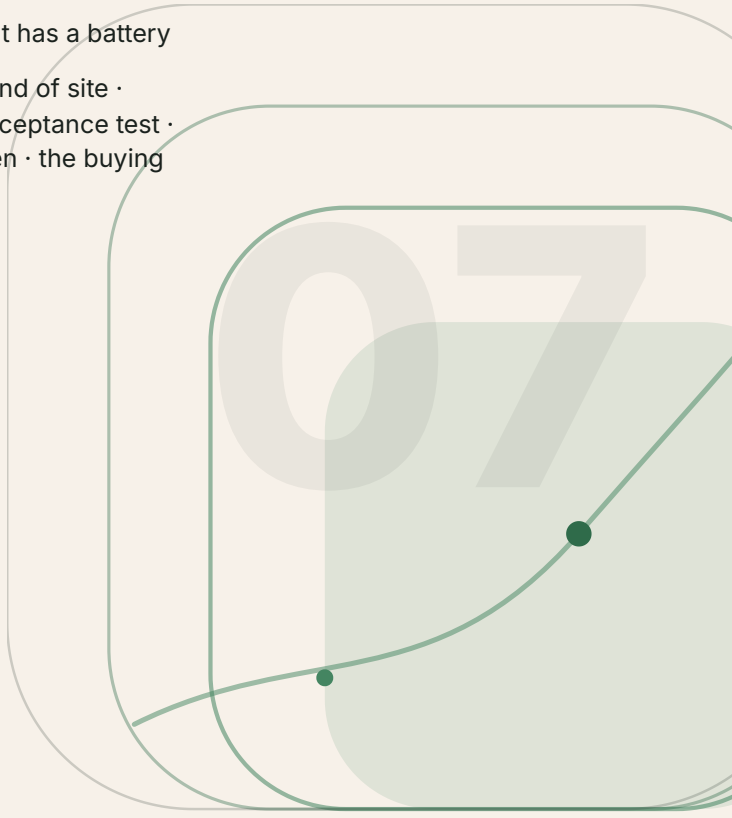
07

HARDWARE SHEET HW07 · QUALIFIED BY ITS REGISTER TABLE

The battery (the customer's own)

A battery at a solar plant is the customer's own equipment. Greecon can read its charge and its power if the maker says where they are.

QUALIFIES	Qualified by its register table. Only if the battery's maker publishes its Modbus register table. Greecon never guesses a register.
USED IN	Optional in kit S (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager at a solar plant that has a battery
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW07

The battery (the customer's own)

A battery at a solar plant is the customer’s own equipment. Greecon buys nothing for it and approves no model. It can be read if, and only if, its maker publishes where its charge and its power are: its Modbus register table. Then its address is entered by hand and its two readings are mapped by hand. If the maker will not publish the table, the battery is left out. Nobody guesses a register.

At a glance

WHAT IT DOES	Reports the battery’s charge (percent) and its power (kW, positive while charging), from the battery maker’s own Modbus registers. It is the customer’s own equipment: Greecon buys nothing for it.
HOW IT QUALIFIES	**Qualified by its register table.** Only if the battery’s maker publishes its Modbus register table. Greecon never guesses a register.
SPECIFICATION	Whatever battery the customer has, provided its maker publishes its Modbus register table. It speaks Modbus on the site’s network, at an address of its own, or on its own RS-485 port
NUMBER ON THE CABLE	Its own, entered by hand (for example 8), if it joins the data cable. None if it is on the network at an address of its own.
FITTED BY	Nobody: it is already there. The project manager enters its address and maps its two readings.
PLANNING ESTIMATE (EUR)	Nothing to buy: it is the customer’s own.
USED IN	Optional in kit S. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ Its maker publishes the battery’s **Modbus register table**, and you have it in your hand: the register of the state of charge and of the power, the kind of number each is, and its scale.
- ☐ It speaks Modbus where the box can reach it: on the site’s network at an address of its own (Modbus TCP), or on its own RS-485 port.
- ☐ On RS-485, it can be set to the cable’s speed (9600 baud, 8 data bits, no parity, 1 stop bit) and to a number no other device has.
- ☐ Reading it needs no password that changes, and no cloud account: the box reads registers and nothing else.

STOP

If the maker will not give the register table, leave the battery out, and say so at handover. A register taken from a forum, or from another model, would show a charge that looks right and is not. Greecon only reads: it never sets a battery's mode, its limits or anything else.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	The hotel template has no battery.
S Solar plant with grid connection	OPTIONAL	0 or 1	The template's Battery system (optional). Two readings are mapped by hand: Battery charge and Battery power. If the plant has no battery, remove the device.
F Farm or greenhouse with irrigation	NOT USED	None	None.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

Reading in Greecon	Unit	What it must be	Standard alert
Battery charge	%	The state of charge, 0 to 100	**Low below 25 percent**, a Check today alert. Set it above the reserve the battery system keeps for itself
Battery power	kW	The battery's power, **positive while charging** and negative while it discharges	None

A charge outside 0 to 100, or a power beyond 2,000 kW either way, is called a reading that cannot be real.

Settings, and where they are made

Nothing is set on the battery by Greecon. What is entered is in the platform, by the project manager, from the maker's table:

What	Where it comes from
Its address	The battery's own address on the network; or, on RS-485, the converter's address
Its port	The maker's table (502 is usual)
Its unit number	The maker's table; on the data cable, a number no other device has (for example 8)
For each reading: the register kind, the address, the kind of number, which half comes first, and the scale	The maker's table

Fitting

Nothing is fitted. The battery is already there, and its installer's work is not touched.

- ☐ If it joins Greecon's data cable (RS-485), PROJECT MANAGER connects A, B and the ground at the battery's communication port, as the battery's manual draws it, at the end of the daisy chain. Nothing inside the battery's power side is opened.
- ☐ If it is on the network, nothing is connected: the box reaches it through the router.

In the platform

The solar template makes the device **Battery system (optional)** with its two readings and no registers.

1. **No battery at this plant:** press **The site does not have this: remove it** under it in step 4, and nothing waits for it.
2. **Its address** (step 4, **Where each device is on the network**): the box cannot recognise a battery on a network, so its line says **Enter by hand**. Under the device open **Enter the address by hand**, fill in the **IP address**, the **Port** and the **Unit number**, and press **Save**.
3. **Its readings** (step 5, **Which register holds which reading**): under each reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx. The address counts from zero (40001 is address 0). Choose the kind of number and, for a number of two registers, which half comes first. Under how it becomes the reading choose **Multiply and add**: a battery that reports watts for a reading in kW is multiplied by 0.001. Press **Save the register**.
4. **Its limit** (step 10, **Alert limits agreed with the customer**): agree the low-charge limit with the owner.

Acceptance test

☐ Each register is read from the box before it is relied on, and the number is the one on the battery's own display:

```
greecon-edge modbus-test --host 192.168.1.60 --unit 8 --type holding --address 0 --data-type uint16 --scale 0.1
```

(The address, the unit, the kind and the scale are the maker's: these are an example.)

Check and how	Pass when	Greecon shows	Real	Result
Battery charge. Against the battery's own display.	The two agree (write both numbers)			☐ Pass ☐ Fail
Battery power. While the battery is charging, and again while it discharges.	Positive while charging, negative while discharging, and the size agrees with the battery's display			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A reading is a thousand times too big	Watts shown as kilowatts	In the register form, multiply by 0.001
The power has the wrong sign (negative while charging)	The maker counts the other way	In the register form, under *Multiply by*, enter a negative number (for example -0.001), so that charging is positive
A reading is constant, or zero	A wrong register, or the wrong register kind (input against holding)	Read the register from the box twice, a minute apart, against the maker's table
The battery does not answer	Its address, its port or its unit number; on RS-485, its speed or A and B	Check each against the maker's table and the battery's own settings page

Not yet proven

- No battery has been read by this software, and no battery model is approved. Every battery is a first time: the acceptance test above is the proof, for that battery.
- The energy picture shows a battery only from these two readings. If only one can be mapped, say at handover which one is missing.

What to write down

There is nothing to buy. Write on the commissioning record:

Field	Answer
Battery make, model and firmware version	
Where the register table came from (document and date)	
Address, port and unit number entered	
Register, kind of number and scale of the charge	
Register, kind of number and scale of the power	
Acceptance test passed on (date), by	